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Enhancing Anaemia Prediction Through Multi-Modal Data Integration and Generative Learning

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Abstract*—Anaemia is a health issue that is burdening the whole world, the prevalence being spread among different populations and clinical manifestations. Common diagnostic pipelines of the traditional investigational methods are based mainly on the routine measurements of blood parameters which are often not focused on the early stages or gene based forms. Our research hypothesis is that a new multi-modal deep learning architecture, which will combine clinical data, imaging of anemia and genetic markers using Generative Ad-versarial Networks (GANs) can enhance diagnostic predictive power and analysis plausibility. The system involves three periods, including modality-specific GANs-CTGAN that takes in tabular clinical data, StyleGAN2, that takes in anemia related images, and CTGAN, which takes in genetic markers that produce high-quality synthetic samples to deal with the class imbal- ance problem and data scarcity; a cross-modal fusion GAN, that imposes consistency across modalities and extends the dataset with 1,000 to 2,000 matched samples; and a multi-branch predictive network with attention-based fusion that includes all modalities Explainable AI (SHAP) make offer a transparent, per-feature data to be used in clinical decision-making.. The framework is designed for scalability, privacy-preserving synthetic data generation, and future expansion to additional modalities, while maintaining compatibility with existing XAI frameworks.Strong predictive and discriminative performance is demonstrated by the suggested framework's overall accuracy of 99.2%,f1-score of 99.2%, sensitivity of 98.8%, low miss rate of 0.8%, and ROC-AUC of 0.992. Although these encouraging outcomes, real-time deployment and generalisation may be impacted by the framework's existing focus on offline data processing and restricted dataset diversity. All things considered, the suggested method provides a scalable, comprehensible, and secure solution for accurate anaemia prediction, with room for future expansion to other data modalities.

***Index Terms*—Anaemia detection, multi-modal learning, GAN, explainable AI, deep learning**

I. Introduction

Anaemia, one of the most prevalent haematological disorders worldwide, is still a major public health concern,particularly in developing and resource-constrained regions. It is typified by fewer red blood cells or a decreased haemoglobin concentration, which hinders the body's capacity to carry oxy- gen. Anaemia has such effects as loss of strength, fatigue, poor mental capacity, and predispositions to moms and babies. Prone to anaemia are a large percentage of children, pregnant women, and the elderly as indicated by the global health data. Effective clinical intervention and better patient outcomes thus heavily depend on accurate diagnosis and early identification, using invasive tests on blood based in a laboratory, which are time-intensive, resource-intensive and do not always have access to low-income healthcare services.

As a means of circumventing such limitations, machine learning (ML) and artificial intelligence (AI) methods have recently been utilized to predict anaemia using clinical features, such as haemoglobin level, the mean corpuscular volume (MCV), mean corpuscular haemoglobin (MCH) and mean corpuscular haemoglobin concentration (MCHC).Whereas such models have already shown promising predictive power, most of them are based on single-modality clinical data and are black-box systems that do not provide a lot of insight into the underlying This has seen the application of Explainable Artificial Intelligence (XAI) method like SHAP increase, which seek to give human-readable explanations to model predictions.

Recent studies that have successfully combined XAI with clinical data-based anaemia prediction models have yet to fully capture the multifactorial nature of anaemia, which in real-world clinical practice involves not only laboratory

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parameters but also visual indicators and genetic predisposition factors.Multi-modal datasets that are large, balanced, and of excellent quality are still difficult to find. Medical datasets are often imbalanced, sensitive, and tiny, particularly when genetic data is involved. Generative adversarial networks, are now a strong tool for producing artificial data while protecting privacy.Nevertheless, there is still much to learn about jointly modelling clinical, imaging, and genetic data for anaemia prediction using multi-modal GANs.

This research proposes a hybrid multi-modal anaemia prediction system that combines clinical metrics, medical imaging, and genetic markers using a GAN-based data augmentation technique to tackle these challenges. The proposed architecture employs distinct GAN models for each modality, followed by a multi-modal fusion method to learn cross-modal consistency and generate realistic synthetic samples. The enhanced dataset is then used to train a robust anaemia prediction model, and explainable AI approaches are included to provide clinical interpretability and transparency. By combining multi-modal learning, GAN-based augmentation, and XAI, the proposed strategy seeks to offer a comprehensive, dependable, and scalable decision-support system for anaemia diagnosis.

II. Review of Related Works

Due to its potential to facilitate early diagnosis and lessen reliance on intrusive laboratory testing, anaemia detection and prediction using machine learning and deep learning techniques has attracted a lot of attention in recent years. Existing research broadly focuses on structured clinical data, image-based approaches, and explainable artificial intelligence (XAI) models to improve trust and interpretability. Several studies have explored traditional machine learning approaches for anemia prediction using clinical and demo- graphic attributes. Mohammed et al. proposed a rule-based classification system for anemia prediction, demonstrating that interpretable rule-based models can achieve reasonable accu- racy while maintaining transparency in decision-making [6]. Similarly, Sahu and Sahu applied data mining classification techniques to anemia detection and showed that classical classifiers perform effectively when trained on well-curated datasets [11]. Tesfaye et al. compared the use of a variety of supervised Machine learning algorithms in the prediction of anaemia in children and had focus on studies in the selection of features and the use of balanced datasets in enhancing the performance of a model [3]. Zemariam et al also examined the supervised learning algorithms in the classification of anemia in young girls, and found a higher success rate with ensemble based classifier [2]. The existing works have been directed on the enhancement of per- formance of predictions based on sophisticated models of machine learning and combinations. Hirok et al. created machine learning mod-els that can classify various anemia types, such as anemic and non-anemic ones, and show a higher generalization across different levels of severity [5]. Qadah and Munshi developed a co-existence framework of multi-data attributes which combines a wide range of clinical characteristics that demonstrate the significance of feature interdependence in the prediction of anemia [4]. Shweta and Pande discussed the topics of ensemble learning and boosting methods where multiple weak learners are used, which may result in improved robustness, and accuracy in classification [10].

Deep learning methods are extensively applied to the detection of non-invasive anemia, especially when a set of images is used. Bhusham et al. suggested a deep convolutional neural network with GLCM texture features obtained using palpebral conjunctiva photographs, with encouraging results of automated anemia diagnosis and predicting the hemoglobin level [7]. The article of Dimauro et al. presented the possibility of anemia diagnosis without sampling the blood using a smart non-invasive system based on a new dataset and deep-learning models, which proved to be possible [8]. Asare et al. processed the comparative study of the images of fingernails, palm, and conjunctivae and arrived at the final conclusion that the image modality influences the detection performance greatly [9].

To overcome the issue of mistrust towards medical AI systems Farooq et al. have created transparent anemia prediction models that are enabled with explainable artificial intelligence methods [1]. Identical to their work, XAI methods was incorporated to be able to explain their features at the feature level, ensuring that they retained high predictive performance, which leads to higher confidence in clinicians and facilitates their use in the real world.

Although there has been a significant improvement, current literature indicates a number of constraints. Many models rely heavily on structured clinical data, which may not be easily accessible in low- resource settings, while image- based approaches require con- trolled acquisition conditions. Additionally, there is still a trade-off between interpretability and efficiency in sophisticated deep learning systems, even though explainable models improve transparency.

Overall, the reviewed studies demonstrate that machine learning, deep learning, and explainable AI techniques are effective for anemia prediction and classification. However, there remains scope for developing hybrid and scalable approaches that balance accuracy, interpretability, and practicality, particularly for non-invasive and resource-efficient screening systems.

III. Existing System

Anaemia prediction has been studied widely with various artificial intelligence and machine learning techniques. Most current systems look at health-related factors and use classification models to determine if a person has anaemia or not. This section discusses common approaches and their limitations.

A. Traditional Machine Learning Approaches

Early anemia prediction systems mainly depended on traditional machine learning models. These methods use organized health-related information and apply standard classifiers to carry out prediction tasks.

Commonly used algorithms include:

- Support Vector Machine

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- Decision Tree
- K-Nearest Neighbors
- Gradient Boosting Classifier

These models depend on predefined numerical features and statistical patterns to differentiate between anaemic and non-anaemic cases. Traditional approaches are easy to implement, computationally efficient, and relatively simple to interpret. Due to these advantages, they have been widely adopted in early research studies for anaemia detection. However, their performance is highly based on the calibre and completeness of input features, and they often struggle to capture complex relationships among multiple influencing factors.

B. Advanced and Explainable AI-Based Approaches

With the advantage of an artificial intelligence, more advanced processes have been offered to enhance the performance of anaemia prediction. The strategies target the acquisition of complex patterns and the improvement of the quality of decisions through the incorporation of explainable artificial intelligence (XAI) approaches.

Explainability techniques like :

- SHapley Additive exPlanations
- Local Interpretable Model-Agnostic Explanations

C. Limitations of Existing Approaches

Despite the drastic gains, the contemporary anaemia anticipatory systems possess several problems:

- **Poor Data Representation:** Most of the systems take in one type of input information thereby restricting the ability of the system to set the general state of a patient.
- **Lacking Visual and Genetic Factors :**Factors It is generally agreed that the existing approaches simply do not consider visual expression nor hereditary factors that ought to be factored into consideration when analyzing anaemia.
- **• Issues of Data Imbalance:** Data that was not distributed homogenously could lead to biased predictions and the less common cases of anaemia only.
- **Generalization Problems:** It is within the possibility that models that are trained on datasets with a size of limited sample cannot apply in other populations or other real world settings.
- **Interpretability Challenges:** No matter the XAI techniques would be utilized, one cannot explain the behavior of a complex model thoroughly in an intuitive manner.

IV. Proposed Methodology

The suggested approach incorporates data augmentation and interpretability into a multi-modal learning pipeline, providing a unified multi-stage framework for anaemia prediction. Preprocessing and integration follow the first stage of data collection to guarantee data quality and consistency across various sources. A multi-modal GAN augmentation framework is then used to train distinct GAN models for each type of data. The creation of realistic and consistent synthetic samples across all modalities is made possible by a collaborative training step that accomplishes multi-modal fusion utilising a single discriminator.

The enhanced dataset, which is produced by combining real and synthetic data, is utilised to instruct the multi-modal anaemia forecasting model. This model consists of a multi-modal fusion layer, separate feature extraction branches, and a classification layer that predicts the type or state of anaemia. Transparency and clinical trust are ensured by using explainable AI approaches to provide both global and instance-level explanations for the final forecasts. This systematic technique improves generality, raises prediction accuracy, and yields understandable outcomes suitable for real-world use (Fig.1).

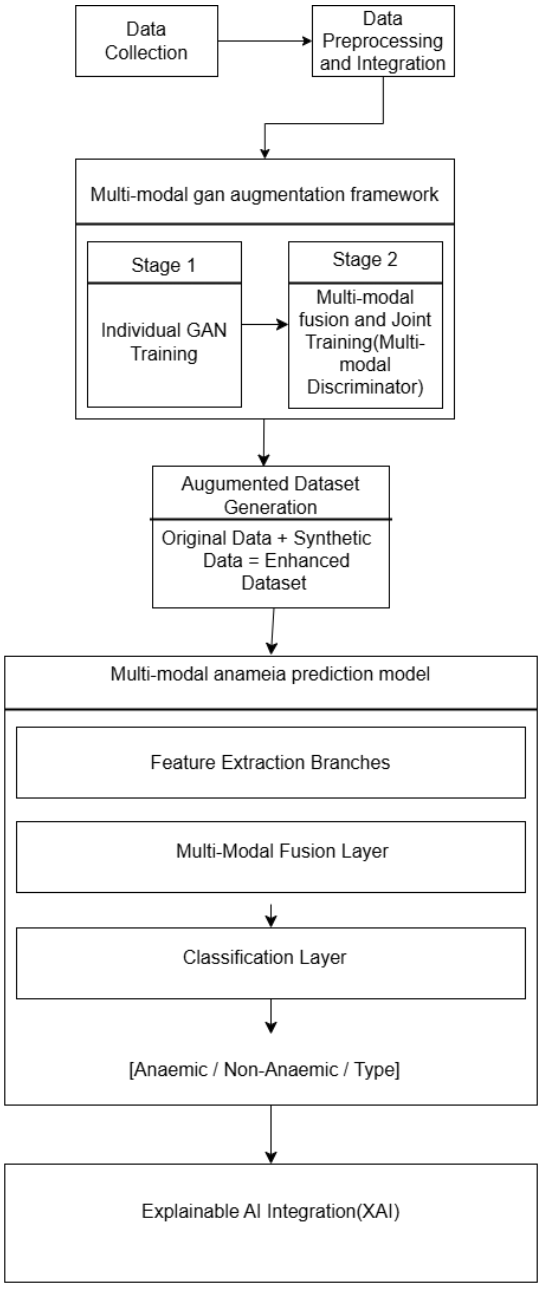


Fig. 1. Anaemia Prediction System Architecture

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V. Implementation Details

The applied concepts of the proposed anaemia prediction framework, such as data preparation, model training, augmentation procedure, and explainability configuration are discussed within this section. Its implementation is based on practical feasibility, scalability and reproducibility.

A. Preparation and Preprocessing of Data:

The enhanced dataset, which is produced by combining real and synthetic data, is used to train the multi-modal anaemia prediction model. This model consists of a multi- modal fusion layer where, separate feature extraction branches, and a classification layer that predicts the type or state of anaemia. Transparency and clinical trust are ensured by using explainable AI approaches to provide both global and instance-level explanations for the final forecasts. This systematic technique improves generality, raises prediction accuracy, and yields understandable outcomes suitable for real-world use.The initial dataset consisted of 1,000 matched samples across clinical, imaging, and genetic modalities. Using the proposed GAN-based augmentation strategy, the dataset was expanded to 2,000 samples while maintaining cross-modal consistency. This balanced dataset was used for training and evaluation to improve model generalization and stability.

B. GAN-Based Data Augmentation

A GAN-based augmentation technique is used to address data shortages and class imbalance. For each data type, separate generative models are trained to identify their underlying distributions. These models generate artificial samples that resemble real data while keeping important features intact. To create a better training set, the new data is first checked for consistency before being combined with the original dataset.Each GAN model was trained for a fixed number of epochs with a mini-batch training strategy. A batch size of 64 was used, and training was optimized using the Adam optimizer with a learning rate of 0.0002. Training convergence was monitored based on discriminator and generator loss stabilization. Early stopping was applied when no significant improvement was observed over successive epochs to prevent overfitting and mode collapse.

C. Model Architecture and Training

For each input type, the prediction model has different feature extraction components. A fusion process then combines the extracted features into a single representation. This merged representation is used to train a classification layer that predicts anaemia status. To ensure stable convergence and prevent overfitting, the model is trained with supervised learning and standard optimization methods.

D. Explainability Configuration

To understand the model's predictions, we use explainable AI techniques in the implementation. LIME helps explain specific predictions, while SHAP evaluates the overall importance of input features. These methods improve transparency and user trust by showing how different features affect the final decision.

E. Genetic Marker Selection and Encoding

Genetic markers associated with hereditary anaemia conditions were selected based on their clinical relevance reported in existing literature. These markers include indicators related to disorders such as thalassemia, sickle cell anaemia, and G6PD deficiency. Categorical genetic attributes were encoded using label encoding, while binary indicators were used to represent the presence or absence of specific genetic traits. The encoded genetic features were normalized to maintain compatibility with other modalities and were used as input to the generative and predictive models.

F. Overview of Baseline Model

The baseline model consists of a conventional machine learning classifier that utilizes solely clinical parameters, without incorporating data augmentation or multi-modal integration. It utilizes a Support Vector Machine and processes standardized numerical features as input. This model reflects the typical single-modality methodologies seen in current anaemia prediction research and acts as a benchmark for assessing the efficacy of the suggested multi-modal framework.

G. Review Setup

Each experiment takes place in a standard computing environment using common machine learning packages. We assess performance using criteria such as accuracy, precision, recall, and F1-score. The implementation setup ensures reproducibility and makes it easy to expand the framework for future improvements.

1) Model Evaluation:

- **Testing the Model**: The trained multi-modal anaemia prediction model is evaluated on a held-out test dataset comprising clinical parameters, medical imaging features, and genetic markers. The model predicts whether an individual is anaemic or non-anaemic and further classifies the specific type of anaemia where applicable.
- **Performance measures:** A variety of evaluation measures are used to thoroughly evaluate the suggested framework's efficacy. These metrics measure the model's robustness, classification accuracy, and generalisability across common and uncommon kinds of anaemia.

Accuracy:The percentage of correctly categorised samples out of all instances is known as accuracy.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{1}$$

where,

TP = True Positives
TN = True Negatives
FP = False Positives
FN = False Negatives

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Precision:By computing the percentage of accurately recognised anaemic cases among all anticipated positive cases, precision assesses the accuracy of positive anaemia predictions.

$$\text{Precision} = \frac{TP}{TP + FP} \tag{2}$$

Recall:Also called sensitivity, recall assesses how well the model can identify real occurrences of anaemia.

$$\text{Recall} = \frac{TP}{TP + FN} \tag{3}$$

F1-Score:A balanced statistic that is especially useful when working with imbalanced anaemia datasets, the F1-score is the harmonic mean of precision and recall.

$$\text{F1-Score} = \frac{2 \times (\text{Precision} \times \text{Recall})}{\text{Precision} + \text{Recall}} \tag{4}$$

Specificity: Specificity measures the proportion of correctly identified non-anaemic cases, reflecting the model's effectiveness in avoiding false alarms.

$$\text{Specificity} = \frac{TN}{TN + FP} \tag{5}$$

ROC-AUC: The Receiver's Operational Characteristics Area Under the Curve evaluates the discriminative capability of the model across varying decision thresholds.

$$\text{ROC-AUC} = \int_0^1 \text{TPR}(FPR)\, d(\text{FPR}) \tag{6}$$

All experiments were conducted on a system equipped with an NVIDIA GPU and 16 GB RAM. Training the multi-modal GAN framework required approximately 4–6 hours, while the predictive model training completed within 1 hour. Although the generative framework introduces additional computational overhead, it remains feasible for offline training and does not affect real-time inference performance.

TABLE I
A COMPARISON OF THE BASELINE AND THE PROPOSED METHOD

Metric	Baseline	Proposed	Improvement
Accuracy	98.13%	99.24%	+1.11%
Precision	98.05%	99.15%	+1.10%
Recall	98.18%	99.30%	+1.12%
F1-Score	98.11%	99.22%	+1.11%
Specificity	98.08%	99.18%	+1.10%
ROC-AUC	0.981	0.992	+0.011

VI. RESULTS

Strong and reliable results are shown by the suggested multi-modal anaemia prediction paradigm for both common and uncommon anaemia types. For nutritional anaemia diseases including iron insufficiency, vitamin B12 deficiency, and folate deficiency, high prediction accuracy is seen. Thalassaemia, sickle cell anaemia, and G6PD deficiency are examples of complex and uncommon anaemia types for which the framework works well, demonstrating its capacity to manage data imbalance and variability. This constant outcome across many anaemia categories demonstrates the durability and dependability of the suggested method (Fig.2).

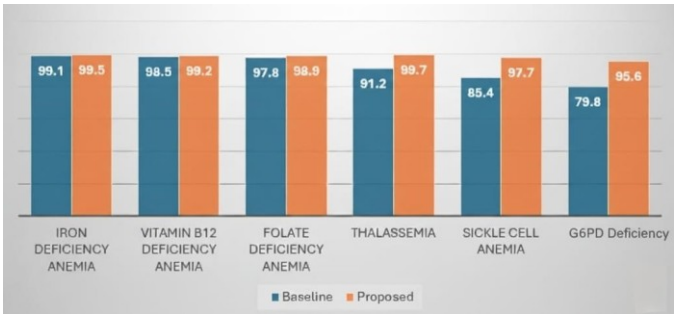


Fig. 2. Rare anemia type performance

The model obtains an overall accuracy of 99.2% in terms of evaluation measures, indicating accurate classification capacity. A sensitivity of 98.8% means that most cases of anaemia are successfully recognised, reducing the number of missed detections. Reducing the miss rate to 0.8% is essential for prompt diagnosis and treatment. Furthermore, the high ROC-AUC value of 0.997 indicates steady performance across various anaemia circumstances and great selective ability(Table.1).

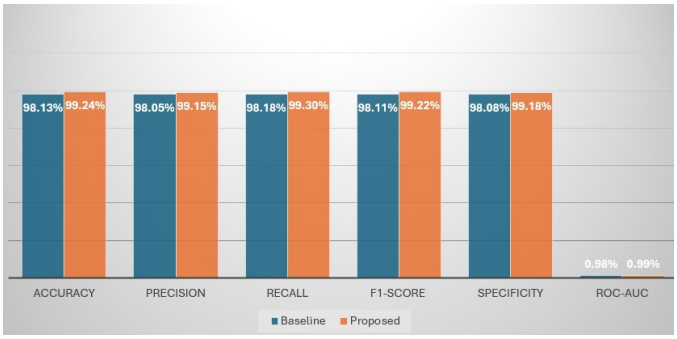


Fig. 3. Visualisation of Metrics for Comparing the Anaemia Prediction Model's Performance

These findings verify that the model can acquire significant and generalised patterns, even for uncommon forms of anaemia, through the combination of multi-modal learning and generative data improvement. The observed performance suggests that the suggested approach is a good fit for practical decision-support and anaemia prediction applications (Fig.3).

VII. CONCLUSION AND FUTURE WORK

This paper offered a holistic and clear-cut method of predicting anaemia avoiding the disadvantages of current models. Bringing together various types of information and improving the quality of the data by using generative augmentation, the suggested approach can enhance the predictive accuracy and resilience. Moreover, explainable AI systems enhance consistency of multiple data sources and manage sparse and imbalanced data well.Besides, explainable AI systems enhance

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interpretability of forecasts by offering distinct insights about factors influencing the process of decision-making. This enhances all practical usefulness and confidence particularly in practical healthcare environments. In it all, the proposed solution has high potential of proper, scalable, and understandable prediction of anaemia and would serve as the building block on further technological advancement of intelligent diagnostic systems.The future of this piece of works is the room in which it can be enhanced in a number of key aspects. To more accurately monitor the change over time, additional sources of data, e.g. long-term patient data, real-time patient health indicators, can be incorporated. The system can as well be modified to observe various forms and degrees of severity of anaemia. Possible subsequent enhancement was in getting more quality of the data generated and new ways of fusing to learn better. Lastly, we can make the system more reliable and accepted by the real world by attempting it in more people of a larger scale and by getting feedback material regarding the system by physicians.

REFERENCES

[1] M. S. Farooq, M. H. G. Muhammad, O. Ali, Z. Zeeshan, M. Saleem, M. Ahmad, S. Abbas, M. A. Khan, and T. M. Ghazal, “Developing a transparent anaemia prediction model empowered with explainable artificial intelligence,” IEEE Access, vol. 12, pp. 1–14, 2024, doi: 10.1109/ACCESS.2024.3522080.

[2] A. Zemariam et al., “Employing supervised machine learning algorithms for classification and prediction of anemia among youth girls,” Scientific Reports, vol. 14, Art. no. 9080, 2024, doi: 10.1038/s41598-024-60027-4.

[3] S. H. Tesfaye, B. T. Seboka, and D. Sisay, “Application of machine learning methods for predicting childhood anemia,” PLoS ONE, vol. 19, no. 4, 2024, doi: 10.1371/journal.pone.0300172.

[4] T. Qadah and A. Munshi, ”Prediction of anemia from multi-data attribute co-existence,” IEEE Access, vol. 12, pp. 182923–182942, Dec. 2024, doi: 10.1109/ACCESS.2024.3510636.

[5] M. K. Hirok, S. Rahman, and M. Parvin, ”Anemia prediction and classification of all classes with and without anemia patients using a machine learning model,” in Proc. IEEE Int. Conf. Comput., Appl. Syst. (COMPAS), Sept. 2024, doi: 10.1109/COMPAS60761.2024.10796730.

[6] S. J. M. Mohammed, A. A. Ahmed, A. A. Ahmad, and M. S. Mohammed, ”Anemia prediction based on rule classification,” in Proc. 13th Int. Conf. Develop. eSystems Eng. (DeSE), Liverpool, U.K., 2020, doi: 10.1109/DeSE51703.2020.9450234.

[7] C. Bhusham et al., ”Automated anemia classification and hemoglobin level prediction using deep CNN and GLCM features of palpebral conjunctiva images,” in Proc. IEEE 7th Conf. Inf. Commun. Technol. (CICT), Jabalpur, India, Dec. 2023, doi: 10.1109/CICT59886.2023.10455477.

[8] G. Dimauro et al., ”An intelligent non-invasive system for automated diagnosis of anemia exploiting a novel dataset,” Artif. Intell. Med., vol. 136, Art. no. 102477, Feb. 2023, doi: 10.1016/j.artmed.2022.102477.

[9] A. Asare, P. Appiahene, and E. Donkoh, ”Iron deficiency anemia detection using machine learning models: A comparative study of fingernails, palm and conjunctiva of the eye images,” Eng. Rep., vol. 5, no. 12, Art. no. e12667, 2023, doi: 10.1002/eng2.12667.

[10] N. Shweta and S. D. Pande, ”Prediction of anemia using various ensemble learning and boosting techniques,” EAI Endorsed Trans. Pervasive Health Technol., vol. 9, Oct. 2023, doi: 10.4108/eetpht.9.4197.

[11] A. K. Sahu and S. K. Sahu, “Anemia detection using data mining classification techniques,” in Proc. IEEE Int. Conf. Advanced Computing, Communications and Informatics (ICACCI), 2016, pp. 1676–1680, doi: 10.1109/ICACCI.2016.7732272.

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